

Data Analytics (DA)

Module-End -3

Module End Assignment:

"Analyzing E-Learning Platform Purchases using MySQL"

Assignment Task Overview

Problem Statement:

An online learning platform sells various digital courses to learners across different countries. Each learner can purchase multiple courses, and each course belongs to a specific category.

The management team wants to analyze purchase data to understand **sales trends**, **learner behavior**, and **popular course categories**.

-- Create Database

```
CREATE DATABASE IF NOT EXISTS elearning_db;
USE elearning_db;
```

-- Table 1: learners

```
create table learners (
  learner_id int primary key auto_increment,
  full_name varchar(100) not null,
  country varchar(50) not null
);
```

-- Table 2: courses

```
create table courses (
  course_id int primary key auto_increment,
  course_name varchar(100) not null,
  category varchar(50) not null,
  unit_price decimal(10, 2) not null
);
```

-- Table 3: purchases

```
create table purchases (
  purchase_id int primary key auto_increment,
  learner_id int not null,
  course_id int not null,
  quantity int not null check (quantity > 0),
  purchase_date date not null,
  foreign key (learner_id) references learners(learner_id),
  foreign key (course_id) references courses(course_id)
);
```

-- Insert Learners

```
insert into learners (full_name, country) values
('John Smith', 'USA'),
('Priya Sharma', 'India'),
('Maria Garcia', 'Spain'),
('David Kim', 'South Korea'),
('Ahmed Hassan', 'UAE');
select * from learners;
```

	learner_id	full_name	country
▶	1	John Smith	USA
	2	Priya Sharma	India
	3	Maria Garcia	Spain
	4	David Kim	South Korea
	5	Ahmed Hassan	UAE
•	NULL	NULL	NULL

-- Insert Courses

insert into courses (course_name, category, unit_price) values

```
('Python for Beginners', 'Programming', 49.99),  
( 'Data Science Masterclass', 'Data Science', 89.99),  
( 'Digital Marketing', 'Marketing', 39.99),  
( 'Web Development', 'Programming', 79.99),  
( 'Data Analytics', 'Data Science', 69.99),  
( 'Share Market', 'Data Science', 69.99);
```

select * from courses;

	course_id	course_name	category	unit_price
▶	1	Python for Beginners	Programming	49.99
	2	Data Science Masterclass	Data Science	89.99
	3	Digital Marketing	Marketing	39.99
	4	Web Development	Programming	79.99
	5	Data Analytics	Data Science	69.99
	6	Share Market	Data Science	69.99
*	NULL	NULL	NULL	NULL

-- Insert Purchases

insert into purchases (learner_id, course_id, quantity, purchase_date) values

-- John Smith (USA)

```
(1, 1, 1, '2026-01-15'),
```

```
(1, 3, 1, '2026-02-10'),
```

-- Priya Sharma (India)

```
(2, 1, 1, '2026-01-20'),
```

```
(2, 2, 1, '2026-01-25'),
```

```
(2, 4, 1, '2026-02-15'),
```

-- Maria Garcia (Spain)

```
(3, 2, 1, '2026-02-01'),
```

```
(3, 5, 1, '2026-02-20'),
```

-- David Kim (South Korea)

```
(4, 4, 1, '2026-01-10'),
```

-- Ahmed Hassan (UAE)

```
(5, 3, 2, '2026-02-28');
```

select * from purchases;

	purchase_id	learner_id	course_id	quantity	purchase_date
▶	1	1	1	1	2026-01-15
	2	1	3	1	2026-02-10
	3	2	1	1	2026-01-20
	4	2	2	1	2026-01-25

DATA EXPLORATION USING JOINS

INNER JOIN:

```
select
    l.full_name as learner_name,l.country,c.course_name,c.category,p.quantity,
    format(c.unit_price, 2) as unit_price,
    format(p.quantity * c.unit_price, 2) as total_revenue,
    date_format(p.purchase_date, '%Y-%m-%d') AS purchase_date
from purchases p
inner join learners l on p.learner_id = l.learner_id
inner join courses c on p.course_id = c.course_id
order by p.purchase_date desc;
```

	learner_name	country	course_name	category	quantity	unit_price	total_revenue	purchase_date
▶	Ahmed Hassan	UAE	Digital Marketing	Marketing	2	39.99	79.98	2026-02-28
	Maria Garcia	Spain	Data Analytics	Data Science	1	69.99	69.99	2026-02-20
	Priya Sharma	India	Web Development	Programming	1	79.99	79.99	2026-02-15
	John Smith	USA	Digital Marketing	Marketing	1	39.99	39.99	2026-02-10
	Maria Garcia	Spain	Data Science Masterclass	Data Science	1	89.99	89.99	2026-02-01
	Priya Sharma	India	Data Science Masterclass	Data Science	1	89.99	89.99	2026-01-25

LEFT JOIN:

```
select
    l.full_name as learner_name,l.country,coalesce(c.course_name, 'No Purchase') as course_name,
    coalesce(p.quantity, 0) as quantity,
    coalesce(format(p.quantity * c.unit_price, 2), '0.00') as total_spent
from learners l
left join purchases p on l.learner_id = p.learner_id
left join courses c on p.course_id = c.course_id
order by total_spent;
```

	learner_name	country	course_name	quantity	total_spent
▶	John Smith	USA	Digital Marketing	1	39.99
	John Smith	USA	Python for Beginners	1	49.99
	Priya Sharma	India	Python for Beginners	1	49.99
	Maria Garcia	Spain	Data Analytics	1	69.99
	Ahmed Hassan	UAE	Digital Marketing	2	79.98
	Priya Sharma	India	Web Development	1	79.99

RIGHT JOIN:

```
select c.course_name,c.category,coalesce(sum(p.quantity), 0) as total_quantity_sold,
    coalesce(format(sum(p.quantity * c.unit_price), 2), '0.00') as total_revenue
from purchases p
right join courses c on p.course_id = c.course_id
group by c.course_id, c.course_name, c.category
order by total_quantity_sold desc;
```

	course_name	category	total_quantity_sold	total_revenue
▶	Digital Marketing	Marketing	3	119.97
	Python for Beginners	Programming	2	99.98
	Data Science Masterclass	Data Science	2	179.98
	Web Development	Programming	2	159.98

ANALYTICAL QUERIES

Q1: Display each learner's total spending along with their country

```
select l.full_name as learner_name,l.country,format(coalesce(sum(p.quantity * c.unit_price), 0), 2) as
total_spending
from learners l
left join purchases p on l.learner_id = p.learner_id
left join courses c on p.course_id = c.course_id
group by l.learner_id, l.full_name, l.country
order by coalesce(sum(p.quantity * c.unit_price), 0) desc;
```

	learner_name	country	total_spending
▶	Priya Sharma	India	219.97
	Maria Garcia	Spain	159.98
	John Smith	USA	89.98
	David Kim	South Korea	79.99
	Ahmed Hassan	UAE	79.98

Q2: Find the top 3 most purchased courses based on total quantity sold

```
select c.course_name,c.category,sum(p.quantity) as total_quantity_sold,
format(sum(p.quantity * c.unit_price), 2) as total_revenue
from courses c
inner join purchases p on c.course_id = p.course_id
group by c.course_id, c.course_name, c.category
order by total_quantity_sold desc
limit 3;
```

	course_name	category	total_quantity_sold	total_revenue
▶	Digital Marketing	Marketing	3	119.97
	Python for Beginners	Programming	2	99.98
	Data Science Masterclass	Data Science	2	179.98

Q3: Show each course category's total revenue and number of unique learners

```
select c.category,
format(sum(p.quantity * c.unit_price), 2) as total_revenue, count(distinct p.learner_id) as
unique_learners
from courses c
inner join purchases p on c.course_id = p.course_id
group by c.category
order by sum(p.quantity * c.unit_price) desc;
```

	category	total_revenue	unique_learners
▶	Programming	259.96	3
	Data Science	249.97	2
	Marketing	119.97	2

Q4: List all learners who have purchased courses from more than one category

```
select
l.full_name as learner_name,l.country,count(distinct c.category) as categories_purchased,c.category
from learners l
inner join purchases p on l.learner_id = p.learner_id
inner join courses c on p.course_id = c.course_id
group by l.learner_id, l.full_name, l.country
having count(distinct c.category) > 1
order by categories_purchased desc;
```

	learner_name	country	categories_purchased	category
▶	John Smith	USA	2	Programming
	Priya Sharma	India	2	Programming

-- Q5: Identify courses that have not been purchased at all

```
select c.course_id,c.course_name,c.category,format(c.unit_price, 2) AS unit_price
from courses c
left join purchases p ON c.course_id = p.course_id
where p.purchase_id is null
order by c.course_name;
```

	course_id	course_name	category	unit_price
▶	6	Share Market	Data Science	69.99

Insights from the E-Learning Database

1. Top Revenue Generating Category

The **Data Science** category generates the highest revenue because multiple learners purchased:

- Data Science Masterclass
- Data Analytics
- Share Market

Insight:

Data Science courses are highly demanded and profitable.

2. Most Popular Courses

Courses with highest purchases:

- Python for Beginners
- Digital Marketing
- Data Science Masterclass

Insight:

Beginner-friendly and career-oriented courses attract more learners.

3. Multi-Category Learners

Some learners purchased courses from more than one category.

Example:

- Priya Sharma purchased:
 - Programming
 - Data Science

Insight:

Cross-category learners are more engaged and valuable customers.

4. Country-Based Customer Trends

Different countries are purchasing courses.

Examples:

- India → multiple purchases
- UAE → higher quantity purchase

Insight:

International learners show strong market potential.

5. Unsold Courses

1 course "Share Market" is courses not purchased at all.

Insight:

These courses may need:

- better marketing
- discounts
- updated content

Data Analysis

1. Descriptive Analysis (What Happened?)

Shows category-wise revenue.

	category	total_revenue
▶	Programming	259.96
	Marketing	119.97
	Data Science	249.97

2. Diagnostic Analysis (Why Happened?)

Programming Category performs better because:

- high-demand skills
- multiple course offerings

3. Predictive Analysis (What may happen in future?)

If more AI/Data Science and programming courses are added, overall revenue may increase.

4. Prescriptive Analysis (What should the business do?"

Create combo offers:

Python + Data Science Masterclass